FINAL PRESENTATION · SESSION 2026–27

AI-Based Real-Time Object Detection and Perception System for Autonomous Vehicles

YOLO11n + BYTETRACK + TTC / ROI + FASTAPI DASHBOARD

Camera-only Mini Project: detect, track, measure distance, fit a dynamic corridor, compute TTC, publish an advisory CONTINUE / WARNING / BRAKE — and never invent ego speed or fake lanes.

VIDEO — YOLO11 — TRACK — DISTANCE — ROI — TTC — DASH

0.418	26.4	10	16
MAP50 · BDD VAL	FPS GPU · RTX 2050	DRIVING CLASSES	TABULATED TESTS

PRESENTED BY · CSE (DATA SCIENCE)

Rupal Sharma 0808DS241115

Sarthak Goyal 0808DS241122

Ritima Rajput 0808DS241113

Suhani Thakur 0808DS241133

GUIDE

Mr. Pankaj Pateriya · Asst. Professor

INSTITUTE

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Pixels are not perception

Urban driving is an open-world visual problem. A box on an image is not enough: a car in the oncoming lane is not the same hazard as a car two metres ahead in the ego path.

SEVEN QUESTIONS EVERY FRAME MUST ANSWER

1. **What** objects are present — class + confidence.
2. **Where** they are — box in the image plane.
3. **Which** of them lie in the driving corridor.
4. **How** they move — identity across frames.
5. **How close** — monocular D = (H · F) / h.
6. **Approaching?** — closing speed and TTC.
7. **Dangerous?** — rule table → CONTINUE / WARNING / BRAKE.

ITEM	IN SCOPE	OUT OF SCOPE
Input	One video or camera	LiDAR, radar, multi-cam
Detection	YOLO11 COCO / BDD	Two-stage / transformers
Tracking	ByteTrack / BoT-SORT	Trained Re-ID
Geometry	Pinhole D + classical ROI	Metric 3D cuboids
Safety	Advisory TTC rules	Actuation, ISO 26262
Software	FastAPI + static UI	Cloud fleet / mobile app

Honesty constraint: if ego speed is not sensed it is published as null. Poor lane evidence is labelled uncertain — never faked.

EIGHT DELIVERED OBJECTIVES

1. Real-time detection of road objects (YOLO11n).
2. Identity tracking across frames (ByteTrack).
3. Corridor perception — in / near / outside / uncertain.
4. Relative motion from consecutive frames.
5. Monocular distance from box height.
6. TTC + four-level risk (SAFE…CRITICAL).
7. Recorded video and live camera input.
8. Nine-page dashboard of every measured number.

Twelve sources, one composition

Four clusters: single-stage detection, driving datasets, multi-object tracking, 3D surveys — plus classical edges, Hough, pinhole geometry and TTC. We do not claim a new detector.

#	SOURCE	APPROACH	KEY RESULT	USE IN AUTOSENSE
1	Redmon et al., CVPR 2016	YOLO one-shot CNN	63.4% mAP · 45 FPS	Real-time backbone
2	Yu et al., CVPR 2020	BDD100K benchmark	24.5 AP · 56.7 MOTA	10-class driving data
3	Zhang et al., ECCV 2022	BYTE association	69.6 MOTA · 71.3 IDF1	Default tracker
4	Mao et al., 3D survey	LiDAR / camera / fusion	BtcDet 78.58% L1	Why we stay 2D
5	Qian et al., PR 2022	Taxonomy of 3D	AP3D / NDS	Depth / fusion gap
6	Bewley et al., SORT	Kalman + Hungarian	Realtime MOT	Tracking-by-detection
7	Wojke et al., DeepSORT	Appearance Re-ID	Fewer ID switches	Identity stability
8	Aharon et al., BoT-SORT	Bag-of-tricks MOT	Strong IDF1	Optional UI tracker
9	Canny, 1986	Optimal edge detector	Standard operator	Dynamic ROI stage
10	Duda & Hart, Hough	Line parameter voting	Robust segments	Lane candidates
11	Ultralytics YOLO11	Train / track / export	YOLO11n 2.59M	Runtime API
12	Lin et al., COCO	80-class protocol	Default names	Bundled yolo11n.pt

> Gap: an inspectable real-time student stack that (i) runs on one camera, (ii) exposes every intermediate, (iii) states uncertainty, (iv) never fabricates ego speed. Composition is the claim.

BDD100K official split, laptop GPU runtime

70,000
TRAIN IMAGES

10,000
VAL IMAGES

20,000
TEST IMAGES

1.29M
TRAIN OBJECTS · 0 INVALID

SPLIT	IMAGES	LABELS	OBJECTS	INVALID
Train	70 000	70 000	1 288 405	0
Validation	10 000	10 000	185 578	0
Test	20 000	20 000	367 728	0

TEN CLASSES · CARS DOMINATE

person · rider · car · bus · truck · bike · motor · traffic light · traffic sign · train · 15 val



TRAIN HOST

Kaggle Tesla T4 · 20 epochs · batch 16 · AMP · imgsz 640 · patience 6 · mosaic closed near end · HSV, ±3°, translate 0.1, scale 0.5, flip 0.5.

RUNTIME HOST · 27 SEP 2026

Windows i5-13420H / RTX 2050 · 540p dashcam @640. GPU 26.4 fps / 30.9 ms. CPU 16.2 fps / 54.4 ms. Boot → first frame 15.7 s.

STACK

Python 3.11 · Ultralytics 8.4 · PyTorch CUDA · OpenCV · NumPy · FastAPI · uvicorn · vanilla HTML/CSS/JS · pytest. Repo **Sarthak156/AutoSense_DR**. Notebook **autonomous-perception**.

04 · IMPLEMENTATION · SEVEN MODULES

One learned model. Geometry and rules for the rest.

VIDEO — YOLO11 — TRACK — DISTANCE — ROI — TTC — DASH

1 · Input
VideoCapture or camera index. Routes /api/source/video|camera. Play / pause / resume / restart / stop. Idle frame: NO INPUT SOURCE.

2 · Detect
YOLO11n track/predict · conf 0.35 · IoU 0.45 · imgsz 640. 182 layers · 2 591 790 params. Swap COCO 80-class or BDD 10-class .pt live.

3 · Track
persist=True · bytetrack.yaml default · botsort.yaml optional. Registry keyed by ID. Disable tracking → TTC unavailable (correct).

4 · Dynamic ROI
Gray → 5×5 blur → Canny → HoughLinesP → RANSAC / quadratic. Reuse ≤ 8 with decay, then fallback trapezoid. Never drops a box.

5 · Distance
D = (H · F) / h · F default 700 px (live-editable). Smooth window 5. Heights: person 1.70 · car 1.50 · truck/bus 3.20 · train 3.50 m.

6 · TTC + Risk + Serve
Closing speed from 8-sample deque. TTC cap 60 s. CRITICAL D≤3 m / 1 s → BRAKE. FastAPI MJPEG + WS ~5 Hz · nine pages.

ROIObjectAssociator returns in / near / outside / uncertain. Safety uses boolean **in_path** only. Overlay HUD draws corridor, boxes, IDs, distance. JPEG quality 80. Config patches deep-merge under a lock.

05 · MODEL TRAINING & EVALUATION

YOLO11n · 20 epochs · val on 10 000 images

0.418	0.227	0.651	0.386	0.721
MAP50 · ALL	MAP50–95	PRECISION	RECALL	CAR MAP50

CLASS	IMAGES	INST.	P	R	MAP50	MAP50-95
all	10 000	185 578	0.651	0.386	0.418	0.227
person	3 220	13 265	0.640	0.433	0.488	0.225
rider	515	649	0.573	0.290	0.312	0.143
car	9 879	102 540	0.702	0.685	0.721	0.434
bus	1 242	1 597	0.623	0.453	0.512	0.386
truck	2 689	4 247	0.633	0.482	0.534	0.379
bike	578	1 007	0.482	0.282	0.290	0.137
motor	334	452	0.620	0.220	0.286	0.132
traffic light	5 653	26 891	0.578	0.529	0.505	0.173
traffic sign	8 221	34 915	0.656	0.491	0.536	0.266
train	14	15	1.000	0.000	0.000	0.000

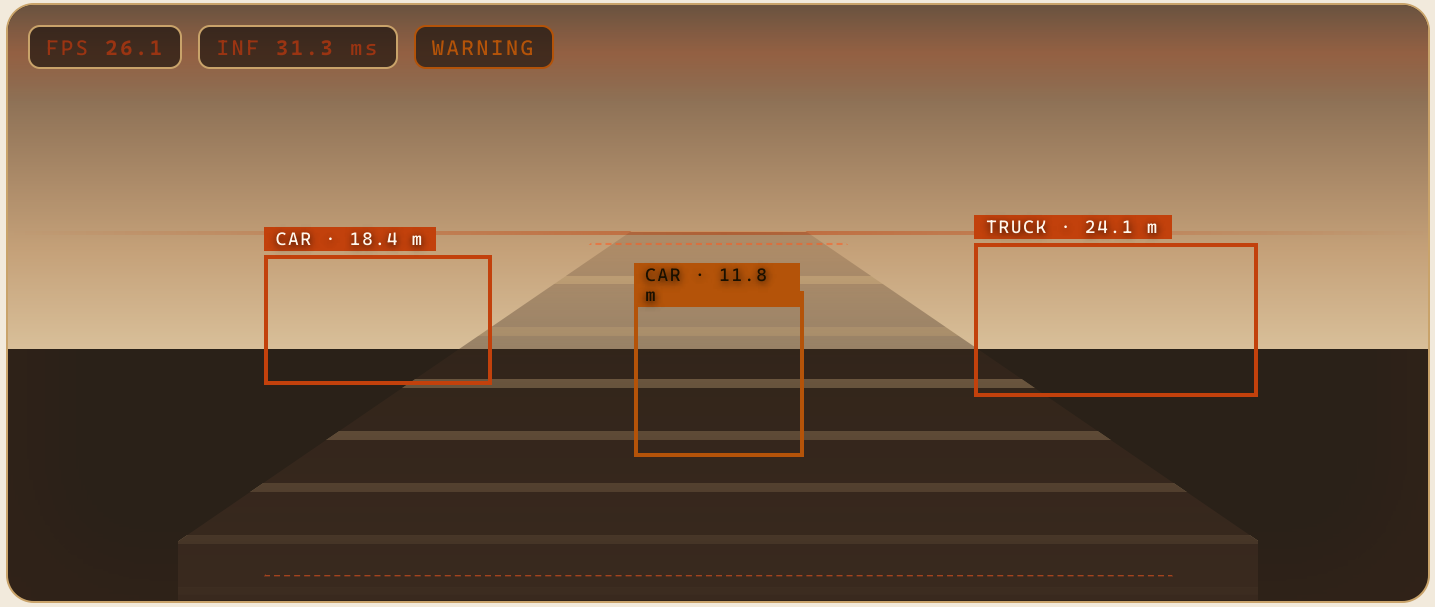
TRAINING RECIPE

COCO-pretrained YOLO11n → BDD100K. 20 epochs · batch 16 · AMP · patience 6 · 182 layers · 2.59M params (2.58M fused). T4: 0.1 ms pre · **1.3 ms inf** · 1.5 ms post. Exported **best.pt** + ONNX 10.1 MB.

HONEST IMBALANCE

102 540 car instances vs 15 train instances. Cars carry the score. Train mAP50 = 0 — reported, not hidden. Extra notebook val/test cells failed on a path mismatch; Table 5.5 is the trainer's own val of best.pt at epoch 20.

Measured on an RTX 2050 — not a slide mock



26.4
FPS GPU 540P

16.2
FPS CPU

30.9
MS GPU INF

54.4
MS CPU INF

CHECK	OBSERVED
GET /healthz	ok true · state running
GET /api/models	yolo11n.pt · 15.9 MB · 80 cls
GET /api/video/frame	JPEG magic ffd8
WS /ws	fps 24.3 on connect
GET /api/report ~9 s	22 frames · 121 det · 23.7 fps
device cpu → auto	16.2 fps ↔ 26.4 fps

Nine pages: Dashboard, Video Input, Detection, Tracking, Distance & ROI, Safety, Visualizations, Model, Settings. Cold sweeps 8–11 fps (GPU power-state). 1080p upload ≈ 7 fps. Contended GPU can collapse inference — benchmark idle.

In-path gating, not raw TTC

CONTINUE
SAFE · not in-path

WARNING
D ≤ 12 m or TTC ≤ 4 s

BRAKE
CRITICAL / DANGER · advisory

LEVEL	DISTANCE	TTC	DECISION
CRITICAL	D ≤ 3.0 m	TTC ≤ 1.0 s	BRAKE
DANGER	D ≤ 6.0 m	TTC ≤ 2.0 s	BRAKE
WARNING	D ≤ 12.0 m	TTC ≤ 4.0 s	WARNING
SAFE	else / not in-path	—	CONTINUE

CLASS	ASSUMED H	CLASS	ASSUMED H
person / rider	1.70 m	car	1.50 m
truck / bus	3.20 m	motorcycle	1.40 m
bicycle	1.50 m	train	3.50 m
traffic light	0.60 m	traffic sign	0.75 m

TRACE A · tunnel_drive.mp4 · imgsz 640

WARNING / WARNING

Closest in-path **11.78 m** (≤ 12 m warning). Min TTC **4.63 s** (above 4 s TTC trigger). Distance rule fired. Not simulated.

TRACE B · dashcam_city.mp4

SAFE / CONTINUE

Min TTC **0.91 s** on an object **outside** the corridor. In-path gating refused BRAKE. ROI never dropped the detection.

Order: not in_path → SAFE; missing D → SAFE; else thresholds. TTC None if closing speed ≤ 0.1 m/s. Advisory only — must not be wired to a vehicle.

V-model + 16 cases — then harden

TC-01 … TC-16 · ALL PASS

TC-01	Curved lane quadratic fit	TC-09	Retain outside boxes
TC-02	Multi-lane centre-prior pair	TC-10	Overlap association
TC-03	Search centre ≠ static trap	TC-11	/healthz running
TC-04	Missing markings → fallback	TC-12	JPEG ffd8 stream
TC-05	Hard shadows uncertain	TC-13	GPU 26.4 fps / 30.9 ms
TC-06	Crosswalk bars rejected	TC-14	CPU 16.2 fps / 54.4 ms
TC-07	Aspect jump clears history	TC-15	In-path WARNING 11.78 m
TC-08	Reuse ≤ 8, then fallback	TC-16	Out-of-path TTC 0.91 s SAFE

pytest without weights (ROI + association + render layers). No k-fold — official BDD split is the substitute. Manual: pause/resume, loop-on-EOF, CAMERA ERROR, MODEL RUNTIME UNAVAILABLE, WS → 350 ms poll.

LIMITS

Inherent

Pinhole D is not LiDAR. Classical ROI fails at night, rain, worn paint, intersections. Risk ignores class and ego speed (null). Train class unlearnable. No auth — localhost only.

NEXT

Harden

JSONL logs, incident stills, optional ego-speed, F calibration wizard, per-class risk, TestClient smokes, ONNX / TensorRT, Dockerfile, config persistence.

NEVER

Non-goals

No actuation. No fake speed. No ADAS / ISO 26262 claim. No stereo/LiDAR in this Mini Project. WARNING/BRAKE are UI labels.

09 · CONCLUSION

From detection to understanding

The Mini Project built a camera-based real-time perception system for research, not a self-driving stack. Existing methods, composed, measured, and discussed without exaggeration.

0.418	0.721	26.4	11.78 m
MAP50 · 10K VAL	CAR MAP50	FPS · RTX 2050	IN-PATH WARNING

WHAT SHIPPED

YOLO11 detect + ByteTrack/BoT-SORT + pinhole distance + training-free corridor + TTC + advisory risk + nine-page FastAPI dashboard. Every FPS, inference_ms, distance and risk label is traceable. Unavailable signals stay unavailable.

SEES → TRACKS → MEASURES → ADVISES

TEAM · CSE (DATA SCIENCE) · 2026-27

Rupal Sharma 0808DS241115 · Sarthak Goyal 0808DS241122
Ritima Rajput 0808DS241113 · Suhani Thakur 0808DS241133

Guide: **Mr. Pankaj Pateriya**, Assistant Professor
IPS Academy, Institute of Engineering & Science, Indore

Questions?

THANK YOU · AUTOSENSE · COMPOSITION IS THE CONTRIBUTION